

STAT 4710/5710: Homework 3

Put your name here

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Instructions

Materials and collaboration

The policy on allowed materials and collaboration is as stated on the Syllabus:

“Students are permitted to work together on homework assignments, but must write up and submit solutions individually. In particular, students may not copy each others’ solutions. Students may consult all course materials, textbooks, the internet, or AI tools (e.g. ChatGPT) to complete their homework. Students may not use solutions to problems that may be available online and/or from past iterations of the course. For each homework, students must disclose all classmates with whom they collaborated, which AI tools they used, and how they used them. Failure to do so will result in a 10-point penalty.”

In accordance with this policy,

Please disclose all classmates with whom you collaborated: None

Please disclose which AI tools you used, and how you used them: ChatGPT in debugging r code and creating visualizations.

Failure to answer the above questions will result in a 10-point penalty.

Writeup

Use `homework-3-template.Rmd` as a starting point for your writeup, adding your solutions after `***Solution***`. Add your R code using code chunks and add your text answers using **bold text**. Consult the [preparing reports guide](#) for guidance on compilation, creation of figures and tables, and presentation quality. If the instructions ask you to “print a tibble”, you should print the tibble directly.

Programming

We encourage you to follow the **tidyverse** paradigm for data visualization, manipulation, and wrangling, as in the first homework, to practice tidyverse workflows (e.g., `filter`, `mutate`, `group_by`, `summarize`, `join`, `ggplot2`).

Students who strongly prefer to use Python for homework may do so. If you choose this option, you are **required** to present both code chunks and the corresponding outputs/interpretations in your PDF. In addition, it’s recommended that you:

- Use **pandas** for data manipulation (preferred, since it parallels `dplyr`).
- Use **plotnine** for visualization (preferred, since it parallels `ggplot2`). If using `seaborn/matplotlib`, you must still include the required plot elements (labels, 45° line, smoothing curve, facets, etc.).

```
library(tidyverse)
library(splines)
library(glmnetUtils)
```

Code completion. In some problems, the task is to complete an incomplete code chunk provided. In those problems, in the template, there is `eval = FALSE` in the chunk header to prevent evaluation errors. You should remove `eval = FALSE` after you have completed the code to enable evaluation and printing of results.

Grading

The point value for each problem sub-part is indicated. There are 100 points possible on this homework, evaluated based on the correctness of the numerical results and plots. When a problem asks for a plot:

- **Axes and labels:** Always include informative axis labels and, if appropriate, titles.
- **Required elements:** If the instructions ask for a 45° line, smoothing curve, or facets, these must be included.
- **Clarity:** Points may be deducted if plots are cluttered, unreadable, or missing labels.

Submission

Compile your writeup to PDF and submit to Gradescope.

1 Case study: logistic regression for home loan (45 points)

1.1 Data import (4 points)

1. Solution.

```
hmeq = read_csv("home_loan.csv")
```

```
## Rows: 3460 Columns: 13
## -- Column specification -----
## Delimiter: ","
## chr (3): REASON, JOB, STATUS
## dbl (10): LOAN, MORTDUE, VALUE, YOJ, DEROG, DELINQ, CLAGE, NINQ, CLNO, DEBTINC
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
hmeq = drop_na(hmeq)
print(hmeq)
```

```
## # A tibble: 1,769 x 13
##   LOAN MORTDUE VALUE REASON JOB YOJ DEROG DELINQ CLAGE NINQ CLNO
##   <dbl> <dbl> <dbl> <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 1700 30548 40320 HomeImp Other 9 0 0 101. 1 8
## 2 1800 28502 43034 HomeImp Other 11 0 0 88.8 0 8
## 3 2400 34863 47471 HomeImp Mgr 12 0 0 70.5 1 21
## 4 2900 7750 67996 HomeImp Other 16 3 0 122. 2 8
## 5 2900 61962 70915 DebtCon Mgr 2 0 0 283. 3 37
## 6 3300 130518 164317 DebtCon Other 9 0 6 192. 0 33
## 7 3700 17857 21144 HomeImp Other 5 0 0 130. 1 9
## 8 3900 29896 45960 HomeImp Other 11 0 0 146. 0 14
## 9 4000 26572 31960 HomeImp Office 11 0 0 118. 0 8
## 10 4400 53091 64074 DebtCon Office 14 1 3 89.9 0 17
## # i 1,759 more rows
## # i 2 more variables: DEBTINC <dbl>, STATUS <chr>
```

2. Solution.

```
hmeq = hmeq |> mutate(STATUS = ifelse(STATUS == "bad", 1, 0))
hmeq$STATUS = factor(hmeq$STATUS, levels = c(0,1))
print(hmeq)
```

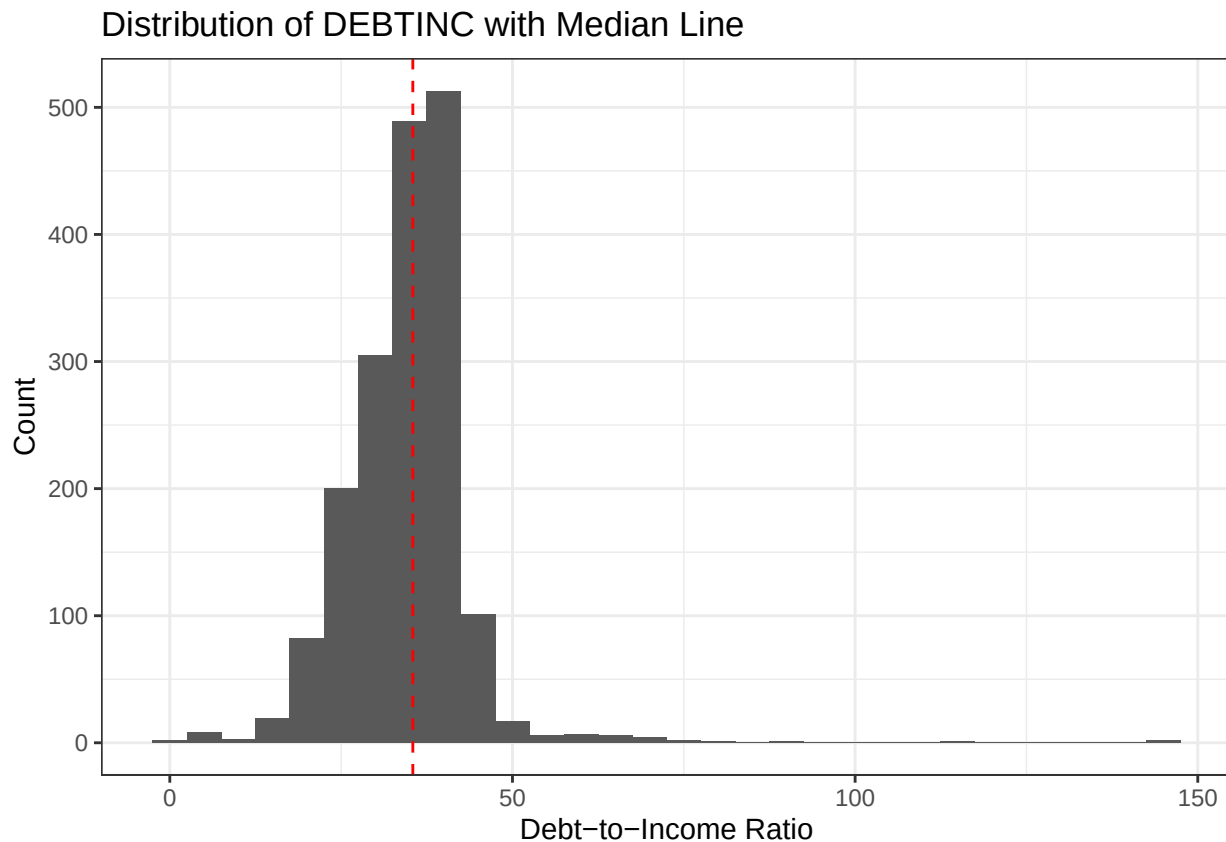
```
## # A tibble: 1,769 x 13
##   LOAN MORTDUE VALUE REASON JOB YOJ DEROG DELINQ CLAGE NINQ CLNO
##   <dbl> <dbl> <dbl> <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 1700 30548 40320 HomeImp Other 9 0 0 101. 1 8
## 2 1800 28502 43034 HomeImp Other 11 0 0 88.8 0 8
## 3 2400 34863 47471 HomeImp Mgr 12 0 0 70.5 1 21
## 4 2900 7750 67996 HomeImp Other 16 3 0 122. 2 8
## 5 2900 61962 70915 DebtCon Mgr 2 0 0 283. 3 37
## 6 3300 130518 164317 DebtCon Other 9 0 6 192. 0 33
```

```
## 7 3700 17857 21144 HomeImp Other 5 0 0 130. 1 9
## 8 3900 29896 45960 HomeImp Other 11 0 0 146. 0 14
## 9 4000 26572 31960 HomeImp Office 11 0 0 118. 0 8
## 10 4400 53091 64074 DebtCon Office 14 1 3 89.9 0 17
## # i 1,759 more rows
## # i 2 more variables: DEBTINC <dbl>, STATUS <fct>
```

1.2 Data exploration and visualization (4 points)

1. Solution.

```
ggplot(data = hmeq, mapping = aes(x=DEBTINC)) +
  geom_histogram(binwidth = 5) +
  geom_vline(aes(xintercept = median(DEBTINC)), color = "red", linetype = "dashed") +
  labs(title = "Distribution of DEBTINC with Median Line",
       x = "Debt-to-Income Ratio",
       y = "Count")
```

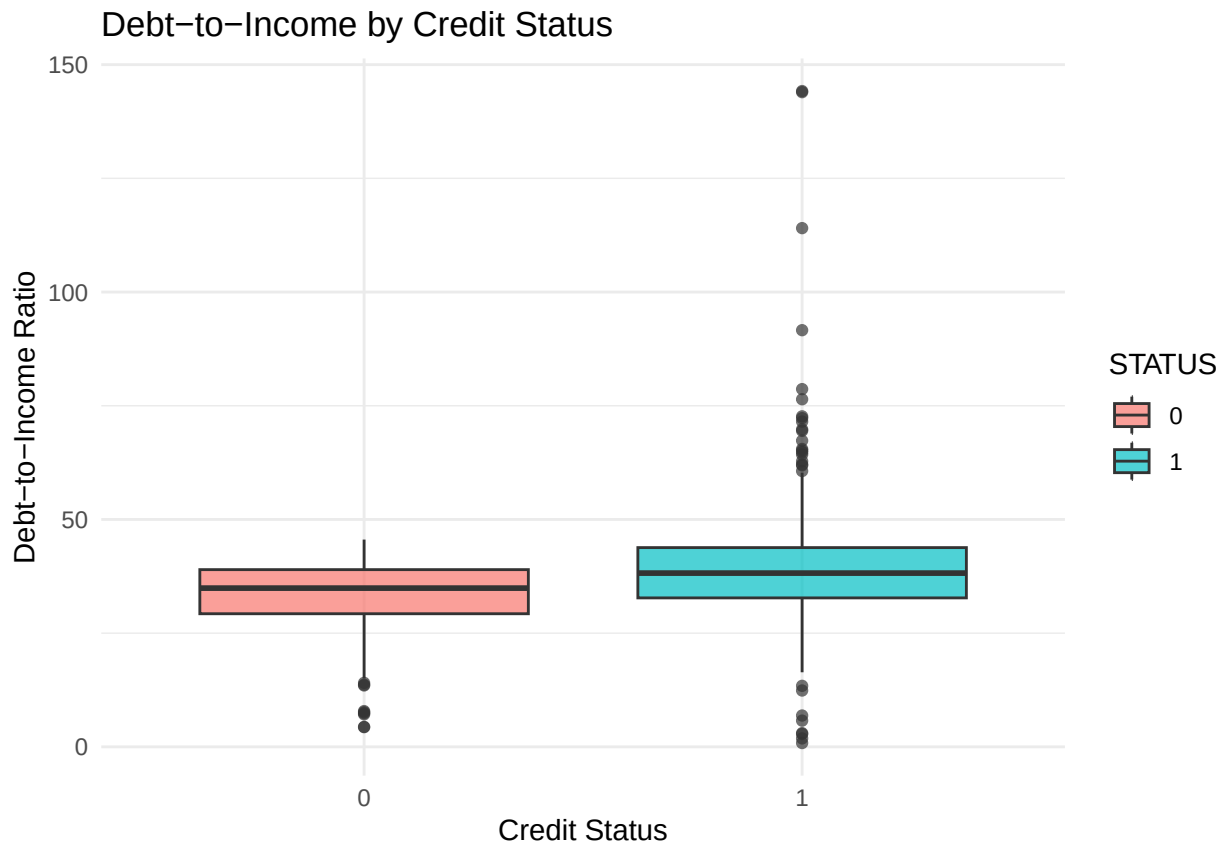


Answer: Based on the histogram, we noticed it's right skewed, where there are some outliers with heavy debt to income ratio.

2. Solution.

```
ggplot(hmeq, aes(x = STATUS, y = DEBTINC, fill = STATUS)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Debt-to-Income by Credit Status",
```

```
x = "Credit Status",
y = "Debt-to-Income Ratio") +
theme_minimal()
```



Answer: we notice the paid loans generally have low debt to income ratio and default loans generally have outliers with high debt to income ratio.

1.3 Univariate logistic regression (23 points)

1.3.1 Understanding logistic regression (13 points)

1. Solution.

```
set.seed(4710) # seed set for reproducibility (DO NOT CHANGE)
n <- nrow(hmeq)
train_samples <- sample(1:n, round(0.8 * n))
hmeq_train = hmeq[train_samples,]
hmeq_test = hmeq[-train_samples,]
```

2. Solution.

```
hmeq_train_subset = hmeq_train |>
  filter(VALUE < 50000 & YOJ == 0 & REASON == "DebtCon" & MORTDUE > 23000) |>
  select(STATUS, DEBTINC)
print(hmeq_train_subset)
```

```
## # A tibble: 4 x 2
##   STATUS DEBTINC
##   <fct>   <dbl>
## 1 1      25.3
## 2 0      16.5
## 3 1      29.3
## 4 0      16.7
```

3. **Solution.** In a logistic regression model, the odds of default are given by

$$\text{odds}(\text{default}) = \exp(\beta_0 + \beta_1 \times \text{DEBTINC})$$

Therefore, the probability of default is

$$p(\text{default}) = \frac{\exp(\beta_0 + \beta_1 \times \text{DEBTINC})}{1 + \exp(\beta_0 + \beta_1 \times \text{DEBTINC})}.$$

Given four observations, with two defaulted and two paid loans, the likelihood function is

$$\begin{aligned} L(\beta_0, \beta_1) &= \prod_{i=1}^4 p_i^{y_i} [1 - p_i]^{1-y_i} \\ &= \left[\frac{\exp(\beta_0 + \beta_1 \times 25.3)}{1 + \exp(\beta_0 + \beta_1 \times 25.3)} \right] \times \left[1 - \frac{\exp(\beta_0 + \beta_1 \times 16.5)}{1 + \exp(\beta_0 + \beta_1 \times 16.5)} \right] \times \\ &\quad \left[\frac{\exp(\beta_0 + \beta_1 \times 29.3)}{1 + \exp(\beta_0 + \beta_1 \times 29.3)} \right] \times \left[1 - \frac{\exp(\beta_0 + \beta_1 \times 16.7)}{1 + \exp(\beta_0 + \beta_1 \times 16.7)} \right]. \end{aligned}$$

4. **Solution.**

```
# fit a logistic model based on the subset training data
fit1 = glm(STATUS~DEBTINC, family="binomial", data=hmeq_train_subset)

# identify beta_0 be -115.38, beta_1 be 5.47
summary(fit1)

##
## Call:
## glm(formula = STATUS ~ DEBTINC, family = "binomial", data = hmeq_train_subset)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -115.38   270307.30      0      1
## DEBTINC         5.47   12308.45      0      1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 5.5452e+00  on 3  degrees of freedom
## Residual deviance: 2.6241e-10  on 2  degrees of freedom
## AIC: 4
##
## Number of Fisher Scoring iterations: 23
```

Answer:

The estimated probability of default, given a debt-to-income ratio x , is

$$P(\text{STATUS} = 1 \mid \text{DEBTINC} = x) = \frac{\exp(-115.38 + 5.47x)}{1 + \exp(-115.38 + 5.47x)}.$$

1.3.2 Univariate logistic regression on the full data (10 points)

1. Solution.
2. Solution.
3. Solution.

1.4 Multivariate logistic regression (14 points)

1. Solution.
2. Solution.
3. Solution.

```
... # TO COMPLETE
misclass_rate <- # TO COMPLETE
precision <- # TO COMPLETE
recall <- # TO COMPLETE
F1 <- # TO COMPLETE

test_errors <- tribble(
  ~Metric, ~Value,
  "Misclassification_rate", misclass_rate,
  "Precision", precision,
  "Recall", recall,
  "F1_score", F1
)
# print the tibble
test_errors
```

2 Case study: linear regression for housing prices (55 points)

2.1 Exploratory data analysis (10 points)

1. Solution.
2. Solution.
3. Solution.

2.2 Predictive modeling preparation (1 point)

Solution.

```
set.seed(4710) # seed set for reproducibility (DO NOT CHANGE)
n <- nrow(ames_data)
train_samples <- sample(1:n, round(0.8 * n))
```

2.3 Ordinary least squares (6 points)

1. Solution.
2. Solution.
3. Solution.

2.4 Ridge regression (13 points)

1. Solution.
2. Solution.
3. Solution.
4. Solution.

2.5 Lasso regression (10 points)

1. Solution.
2. Solution.
3. Solution.

2.6 Evaluation and comparison (15 points)

1. Solution.

```
... # TO COMPLETE
mse_ols <- ... # TO COMPLETE
mse_ridge <- ... # TO COMPLETE
mse_lasso <- ... # TO COMPLETE

error_for_models <- tribble(
  ~Model, ~RMSE,
  "Linear", mse_ols,
  "Ridge", mse_ridge,
```



```
  "Lasso", mse_lasso
)
# print the tibble
error_for_models
```

2. **Solution.**
3. **Solution.**
4. **Solution.**